The time shift between Mars and Earth, the calculus equation of time

The time shift between two synchronized watches is related to the distance from the Sun of the planets Mars and Earth. The Sun is the principal generator of Tau neutrinos; thus, the time in our neighborhood.

For the time flow Flux Tau neutrino, we have the second order (wave equation) $\frac{\partial^2 P(r,t)}{c^2 \partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = 0$ (1)

With initial condition P(R,t)=0 & P(r0,t) = 0. Where r_0 is ≈ 3.8 · $R_{Schwarzschild}$ of the Sun, the threshold for a Black Hole (11225.35 m) $\boxed{P(r,t) = A\,[(r-r_0)(r-R) + \varepsilon\, t + c^2 t^2]}$

(2)

$$P(r,t) = Ar^2 - A(r_0 + \mathrm{R})\mathrm{r} - \mathrm{A}r_0 R + \mathrm{A}\varepsilon\, t + \mathrm{A}c^2 t^2$$

And for $t_0 = 0$, the flux is proportional to: $P(r,0) = 1.6849 \times 10^{-29}(r-r_0)(r-R)$

The initial temporal rate is linked to the Earth's time quantum $\tau_q = 1.765 \times 10^{-19}$ s : => $\frac{\partial P}{\partial t}\Big|_{t=0} = A \cdot \varepsilon = \tau_q\, c^2$

$A \cdot \varepsilon = 1.765 \times 10^{-19} \times (2.998 \times 10^8)^2 = 1.5863 \times 10^{-2}$ m²/s

$$r_0 = 11222.35 \text{ m}, \quad R = 4.2355 \times 10^{22} \text{ m}$$

$$\boxed{\varepsilon = 1.6610 \times 10^{15} \text{ m}^2/\text{s}}$$

$$\boxed{A = 9.5510 \times 10^{-18}/\text{m}^2}$$

Time flow ~ flux/ surface unit P(r,t) /4πr²

The perception of time on Mars is slower than on Earth, and considering the initial value of time t = 0, and the pas slower on Mars.

According to https://michaelvio.github.io/Tau.pdf => With a solution for null flux r_0 = 11222.35 m, thus the Temporal Event Horizon is at 11.222 Km.

Link PDF format, and https://michaelvio.github.io/GFtau.pdf

Thus, each day on Mars, time loses ~ 5.36 milliseconds/day, and time is slow on Mars, depending on the distance to the Sun (249 million km or 206 million km). The time shift is detectable with an atomic watch.
The flux increment relative to the Local Universe with r in meters distance to the Sun.
Thus, each second on Mars time loses, and time is slow due to the Tau neutrino, depending on the distance to the Sun (249 million km or 206 million km).
Still, the gravitational relativistic effect is weaker on Mars, and the total effect is 477 μs per Earth Day.
https://www.universetoday.com/articles/do-you-know-what-time-it-is-if-youre-on-mars-now-you-do
The time shift is detectable with an atomic watch. The flux increments relative to r in meters from the Sun.
But also, the Earth has a variable distance to the Sun between 147 and 152 million km, thus a few nanoseconds variation. Surprisingly, the GPS satellite is 20000km in the sky between the two positions in front of the Sun on the day and the night towards the Sun [2· (6371+20000)Km]=> ≈ ± 1.35 μs/hr.
https://michaelvio.github.io/MarsGeo.pdf
For geostationary satellites, [2· (6371+35800)Km], the drift in time flow per hour is ≈ 2,18 μs/hr or ≈ 51 μs/day due to gravity and (+ 0÷0.2 μs/hr Sun-facing & − 0÷0.2 μs/hr anti-Sun) ± 4.36 μs per day due to tau neutrino

10 June 2026